furthest; in the example, P_3 would be the next one to be processed.

When we were discussing DBSCAN, we insisted about the order in which points are processed: for DBSCAN, the only thing that matters is that points in the same cluster (i.e. core points reachable from each other) are processed together; as we have seen, however, this is more a matter of optimization, so that we don't have to keep a disjoint set and merge clusters when we find pair of directly reachable points, because the order of processing may only influence the assignment of non-core points on the edge of clusters.

The idea behind OPTICS is that, instead, this order does matter, and in particular it can make sense to keep expanding a "frontier" for current cluster by adding the unprocessed point that is closest to the cluster (if it is reachable from the cluster).

Figure 12.15 illustrates this concept in a simplified scenario. In order to choose the right point, it's obviously necessary to keep track of the distances of all undiscovered neighbors of a cluster[20], and for this purpose the authors of the paper introduced two definitions: to each point, they associated a *core distance*, and a *reachability distance*.

12.4.1 Definitions

The core distance of a point p is the minimum distance for that point to be considered a core point; since the definition of core point, given in section 12.3 for DBSCAN, depends on the parameters ε, the radius of dense regions, and